Unit 09: Ratio and Proportion

CONTENTS

Objectives

The key concepts explained in this unit, including area a learner is expected to learn and master after going through the unit are: -

- We will understand how to comparison of two quantities
- We will understand how to find Proportion
- We will understand how to Compare in a given mixture
- We will understand how to find Fraction
- We will understand how to find Percent in a given mixture
- We will understand how to find Equivalent ratios
- We will understand how to find Mean Proportional

Introduction

A ratio is a comparison of two quantities by division. It is a relation that one quantity bears to another with respect to magnitude. In other words, ratio means what part one quantity is of another. The quantities may be of same kind or different kinds. For example, when we consider the ratio of the weight 45 Kg of a bag of rice to the weight 29 Kg of a bag of sugar, we are considering the quantities of same kind but when we talk of allotting 2 cricket bats to 5 sportsmen, we are considering quantities of different kinds. Normally, we consider the ratio between quantities of the same kind.

If a and b are two numbers, the ratio of a to b is a/b or a ÷ b and is denoted by a:b. The two quantities that are being compared are called terms. The fi rst is called antecedent and the second term is called consequent.

For example, the ratio 3:5 represents 3/5 with antecedent 3 and consequent 5.

1. A ratio is a number in order to find the ratio of two quantities and they must be expressed in the same units.

2. A ratio does not change, if both of its terms are multiplied or divided by the same number. Thus, $\frac{2}{3} = \frac{4}{6} = \frac{6}{9}$.

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9.1 Ratios and Proportions

At a certain college, there are 500 students who are under 25 years of age and 1500 students who are 25 years old or older. The ratio of students under 25 to students over 25 is 500 to 1500, which is also written 500:1500 or as a fraction $\frac{500}{1500}$. We know that as fractions $\frac{500}{1500}=\frac{1}{3}=\frac{1500}{4500}$. A ratio is a division relationship between two numbers. The ratio of a to b, written as a:b, is another meaning for the division or fraction $\frac{a}{b}$.

Equivalent ratios: We have already noticed that the ratios $\frac{500}{1500}=\frac{1}{3}$ are equal as fractions. Notice too that $500 \cdot 3 = 1 \cdot 1500$. Two ratios are equal if and only if their cross products are equal. $\frac{a}{b}=\frac{c}{d}$ if and only if $a \cdot d = c \cdot b$. Observe how the cross products are constructed. The numerator on the left side times the denominator on the right is equal to the numerator on the right times the denominator on the left.

Proportions: An equation of two ratios is called a proportion. $\frac{500}{1500}=\frac{1}{3}$ is a proportion, and $\frac{64}{360}=\frac{16}{90}$ is a proportion. The equality of two ratios is called proportion. If a/b = c/d , then a, b, c and d are said to be in proportion and we write a:b::c:d. This is read as "a is to b as c is to d".

A proportion involves 4 numbers. We can use cross-multiplication and some algebra to solve a proportion equation for an unknown.

Example: Find t so that this proportion is true.

Set the cross products equal and solve the equation.

$$\frac{140}{210} = \frac{8}{t}$$
$$140t = 8 \cdot 210$$
$$140t = 1680$$
$$t = \frac{1680}{140}$$
$$t = 12$$

Proportions can be used to solve many word problems.

Example 2: How far does a train going 68 miles per hour travel in 3 and a half hours?

$$\frac{68 \text{ miles}}{1 \text{ hour}} = \frac{x \text{ miles}}{3.5 \text{ hours}}$$
$$68 \cdot 3.5 = x \cdot 1$$
$$x = 238 \text{ miles}$$

Example: 200 is 80% of what number?

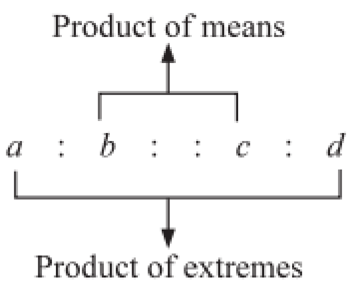
Let b represent the unknown number which is the base in the percent relationship and set up the proportion. Cross-multiply and solve. It helps to reduce the ratio first.

$$\frac{200}{b} = \frac{80}{100}$$
$$\frac{200}{b} = \frac{4}{5}$$
$$4b = 200 \cdot 5$$
$$b = \frac{1000}{4} = 250$$

Exercises:

For example, since 3/4 = 6/8 , we write 3:4::6:8 and say 3, 4, 6 and 8 are in proportion. Each term of the ratio a/b and c/d is called a proportional. a, b, c and d are respectively the fi rst, second, third and fourth proportionals. Here, a, d are known as extremes and b, c are known as means.

1. If four quantities are in proportion, then Product of Means = Product of Extremes For example, in the proportion a:b::c:d, we have, bc = ad



From this relation, we see that if any three of the four quantities are given, then the fourth quantity can be determined.

Fourth proportional: If a:b::c:x, then x is called the fourth proportional of a, b, c.

We have a/b = c / x or x = b*c/x Thus, fourth proportional of a, b, c is b * c/ a

2:5::4:x or, 2/ 5 = 4/x

∴ x = $\frac{5 \times 4}{2}$ = 10

Third Proportional: If a:b::b:x, then x is called the third proportional of a, b.

We have , a/b = b/ x or $b^2/2$ Thus, third proportional of a, b is b^2/a

Example: Find a third proportional to the numbers 2.5, 1.5.

Solution: Let x be the third proportional, then

2·5:1·5::1·5:x or $\frac{2.5}{1.5} = \frac{1.5}{x}$ ∴ $x = \frac{1.5 \, X \, 1.5}{2.5}$ = 0.9

Mean Proportional: If a:x::x:b, then x is called the mean or second proportional of a, b. We have, a x = b x or, x^2 = ab or, x = $\sqrt{ab}$ ∴ Mean proportional of a and b is $\sqrt{ab}$. We also say that, a, x, b are in continued proportion.

Example: Find the mean proportional between 48 and 12.

solution :- let x be the mean proportional . Then

48:x::x:12 or $\frac{48}{x} = \frac{x}{12}$ or x^2 = 576 or x = 24

$\frac{a+b}{b} = \frac{c+d}{d}$ (componends)

1) $\frac{a+b}{b} = \frac{c+d}{d}$ (componends)

2) $\frac{a-b}{b} = \frac{c-d}{d}$ (dividendo)

3) $\frac{a\mp b}{a-b} = \frac{c-d}{c-d}$ (compound and dividend)

4) $\frac{a}{b} = \frac{a+c}{b+d} = \frac{a-c}{b-d}$

Example: Find a third proportional to the numbers 2.5, 1.5.

Solution: Let x be the third proportional, then

2·5:1·5::1·5:x or $\frac{2.5}{1.5} = \frac{1.5}{x}$ ∴ $x = \frac{1.5 \, X \, 1.5}{2.5}$ = 0.9

Mean Proportional: If a:x::x:b, then x is called the mean or second proportional of a, b. We have, a x = b x or, x^2 = ab or, x = $\sqrt{ab}$ ∴ Mean proportional of a and b is $\sqrt{ab}$. We also say that, a, x, b are in continued proportion.

Example: The sum of two numbers is c and their quotient is $\frac{p}{q}$. Find the numbers.

Solution: Let the numbers be x, y.

Given: x + y = c …(1)

And $\frac{x}{y} = \frac{p}{q}$ …(2)

∴ $\frac{x}{x+y} = \frac{p}{p+q}$

= $\frac{x}{c} = \frac{p}{p+q}$ [using (1)]

= x = $\frac{pc}{p+q}$

SHORTCUT – METHOD

a) If two numbers are in the ratio of a:b and the sum of these numbers is x, then these numbers will be $\frac{ax}{a+b}$ and $\frac{bx}{a+b}$, respectively.

Or

If in a mixture of x litres, two liquids A and B are in the ratio of a:b, then the quantities of liquids A and B in the mixture will be $\frac{ax}{a+b}$ litres and bx a $\frac{bx}{a+b}$ litres, respectively.

b) If three numbers are in the ratio of a:b:c and the sum of these numbers is x, then these numbers will be $\frac{ax}{a+b+c}$, $\frac{bx}{a+b+c}$ and $\frac{cx}{a+b+c}$ respectively

SOLUTION:

Let the three numbers in the ratio a:b:c be A, B and C.

Then, A = ka, B = kb, C = kc and, A + B + C = ka + kb + kc = x

⇒ k(a + b + c) = x

⇒ k = $\frac{x}{a+b+c}$

A = ka = $\frac{ax}{a+b+c}$

B = kb= $\frac{bx}{a+b+c}$

C = kc =. $\frac{cx}{a+b+c}$

Example: Two numbers are in the ratio of 4:5 and the sum of these numbers is 27. Find the two numbers. Solution: Here, a = 4, b = 5 and x = 27.

∴ the first number = $\frac{ax}{a+b} = \frac{4\ x\ 27}{4+5}$ = 12

And , the Second number = $\frac{bx}{a+b} = \frac{5\ x\ 27}{4+5}$ = 15

Example: Three numbers are in the ratio of 3:4:8 and the sum of these numbers is 975. Find the three numbers. Solution: Here, a = 3, b = 4, c = 8 and x = 975.

∴ the first number = $\frac{ax}{a+b+c} = \frac{3x\ 975}{3+4+8}$ = 195

And , the Second number = $\frac{bx}{a+b+c} = \frac{4x\ 975}{3+4+8}$ = 260

and, the third number = $\frac{cx}{a+b+c} = \frac{8\ x\ 975}{3+4+8}$ = 520

2) If two numbers are in the ratio of a:b and difference between these numbers is x, then these numbers will be

a) $\frac{ax}{a-b}$ and $\frac{bx}{a-b}$, respectively (where a>b)

b) $\frac{ax}{b-a}$ and $\frac{bx}{b-a}$, respectively (where a<b)

Solution

Let the two numbers be ak and bk. Let a > b. Given: ak - bk = x

⇒ (a - b)k = x or, k = $\frac{ax}{b-a}$

Therefore, the two numbers are = $\frac{ax}{b-a}$ = $\frac{bx}{b-a}$

Example 10: If A:B = 3:4 and B:C = 8:9, find A:B:C.

Solution: Here, $n_1 = 3$, $n_2 = 8$, $d_1 = 4$ and $d_2 = 9$.

∴ A:B:C = $(n_1 \times n_2):(d_1 \times n_2):(d_1 \times d_2)$

= (3 × 8):(4 × 8):(4 × 9)

= 24:32:36 or, 6:8:9.

Example 11: If A:B = 2:3, B:C = 4:5 and C:D = 6:7, find A:D.

Solution: Here, $n_1 = 2$, $n_2 = 4$, $n_3 = 6$, $d_1 = 3$, $d_2 = 5$ and $d_3 = 7$.

∴ A:B:C:D = $(n_1 \times n_2 \times n_3):(d_1 \times n_2 \times n_3):(d_1 \times d_2{:}n_3):(d_1 \times d_2 \times d_3)$

= (2 × 4 × 6) :(3 × 4 × 6):(3 × 5 × 6) :(3 × 5 × 7)

= 48:72:90:105 or, 16:24:30:35.

Thus, A:D = 16:35.

4) (a) The ratio between two numbers is a:b. If x is added to each of these numbers, the ratio becomes c:d. The two numbers are given as:

$\frac{ax(c-d)}{ad-bc}$ and $\frac{bx(c-d)}{ad-bc}$

Solution :-

Let two number be ak and bk.

Given : $\frac{ak+x}{bk+x} = \frac{c}{d} \rightarrow$ akd + dx = cbk + cx

= k(ad-bc) = x(c-d)

= k = $\frac{x(c-d)}{ad-bc}$

Therefore , the two number are $\frac{ax(c-d)}{ad-bc}$ and $\frac{bx(c-d)}{ad-bc}$

(b) The ratio between two numbers is a:b. If x is subtracted from each of these numbers, the ratio becomes c:d.

The two numbers are given as: $\frac{ax(d-c)}{ad-bc}$ and $\frac{bx(d-c)}{ad-bc}$

Solution:

Let two number be ak and bk.

Given : $\frac{ak+x}{bk+x} = \frac{c}{d} \rightarrow$ akd -xd = bck -xc

= k(ad-bc) = x(d-c)

= k = $\frac{x(d-c)}{ad-bc}$

Therefore, the two number are $\frac{ax(d-c)}{ad-bc}$ and $\frac{bx(d-c)}{ad-bc}$

Example 12: Given two numbers which are in the ratio of 3:4. If 8 is added to each of them, their ratio is changed to 5:6. Find the two numbers.

Solution: We have, a:b = 3:4, c:d = 5:6 and x = 8.

The first number $\frac{ax(c-d)}{ad-bc}$

= $\frac{3\ x\ 8\ x\ (5-6)}{(3\ x\ 6\ -4x\ 5)}$ = 12

The second number $\frac{bx(c-d)}{ad-bc}$

= $\frac{5\ x\ 8\ x\ (5-6)}{(3\ x\ 6\ -4x\ 5)}$ = 16

Example 13: The ratio of two numbers is 5:9. If each number is decreased by 5, the ratio becomes 5:11. Find the numbers.

Solution: We have, a:b = 5:9, c:d = 5:11 and x = 5

The first number $\frac{ax(c-d)}{ad-bc}$

$= \frac{5\,x\,5\,x\,(11-15)}{(5x11\ -9x\ 5)} = 16$

The second number $\frac{bx(c-d)}{ad-bc}$

$= \frac{9\,x\,5\,x\,(11-15)}{(5x11\ -9x\ 5)} = 27$

5) (a) If the ratio of two numbers is a:b, then the numbers that should be added to each of the numbers in order to make this ratio c:d is given by $\frac{ad-bc}{c-d}$.

Solution :-

Let the required number be x.

Given: $\frac{a+x}{b+x} = \frac{c}{d}$

= ad + xd = bc + xc

=x(d-c = bc – ad

Or , x = $\frac{ad-bc}{c-d}$

(b) If the ratio of two numbers is a:b, then the number that should be subtracted from each of the numbers in order to make this ratio c:d is given by$\frac{bc-ad}{c-d}$

Solution :-

Let the required number be x.

Given : $\frac{a-x}{b-x} = \frac{c}{d}$

= ad - xd = bc - xc

=x(c-d) = bc – ad

Or , x =$\frac{bc-ad}{c-d}$

Example: Find the number that must be subtracted from the terms of the ratio 5:6 to make it equal to 2:3.

Solution: We have, a:b = 5:6 and c:d = 2:3.

∴ the required number = $\frac{bc-ad}{c-d}$

$= \frac{6\,x2\ -5\,x3}{2-3} = 3$

Example 15: Find the number that must be added to the terms of the ratio 11:29 to make it equal to 11:20.

Solution: We have, a:b = 11:29 and c:d = 11:20.

∴ $The\ required\ number = \frac{ad-bc}{c-d}$

$= \frac{11\,x\,20\ -29\,x\,11}{11-20} = 11$

6) There are four numbers a, b, c and d.

(i) The number that should be subtracted from each of these numbers so that the remaining numbers may be proportional is given by

$$\frac{ad - bc}{(a+d)-(b-c)}$$

Solution

Let x be subtracted from each of the numbers.

The remainders are a - x, b - x, c - x and d - x.

Given: $\frac{a-x}{b-x} = \frac{c-x}{d-x}$

⇒ (a - x) (d - x) = (b - x) (c - x)

⇒ ad - x(a + d) + x^2 = bc - x(b + c) + x^2

⇒ (b + c)x - (a + d)x = bc – ad

∴ x = $\frac{bc - ad}{(b+c)-(a+d)}$ or $\frac{ad - bc}{(a+d)-(b+c)}$

(ii) The number that should be added to each of these numbers so that the new numbers ma be proportional is given by

x = $\frac{bc - ad}{(a+d)-(b+c)}$

Solution

Let x be added to each of the numbers. The new numbers are a + x, b + x, c + x and d + x.

Given : $\frac{a+x}{b+x} = \frac{c+x}{d+x}$

⇒ (a + x) (d + x) = (b + x) (c + x)

⇒ ad + x(a + d) + x^2 = bc + x(b + c) + x^2

⇒ (a + d)x - (b + c)x = bc - ad.

∴ x = $\frac{bc - ad}{(a+d)-(b+c)}$

Example 16: Find the number subtracted from each of the numbers 54, 71, 75 and 99 leaves the remainders which are proportional. Solution: We have, a = 54, b = 71, c = 75 and d = 99

The required number = ∴ x = $\frac{ad-bc}{(a+d)-(b+c)}$

= $\frac{54 \times 99 - 71 \times 75}{(54+99)-(71+75)}$ = 3

7) The incomes of two persons are in the ratio of a:b and their expenditures are in the ratio of c:d. If the saving of each person be `S, then their incomes are given by

₹$\frac{aS(d-c)}{ad-bc}$ and ₹$\frac{bS(d-c)}{ad-bc}$

and their expenditures are given by

₹$\frac{cS(d-a)}{ad-bc}$ and ₹$\frac{aS(d-a)}{ad-bc}$

Solution

Let their incomes be ₹ ak and ₹ bk, respectively. Since each person saves ₹ S,

∴ Expenditure of fi rst person = ₹(ak - S)

and, expenditure of second person = ₹(bk - S).

Given $\frac{ak-S}{bk-S} = \frac{c}{d}$

= akd - Sd = bkc - Sc

⇒ k(ad - bc) = (d - c)S or, k = $\frac{(d-c)}{ad-bc}$

Therefore, the incomes of two persons are

$\frac{a(d-c)S}{ad-bc}$ and ₹ $\frac{b(d-c)S}{ad-bc}$

And their expenditure are

Ak-S and bk-s

i .e $\frac{a(d-c)S}{ad-bc}$ -S and ₹ $\frac{b(d-c)S}{ad-bc}$-S

$\frac{aS(d-c)}{ad-bc}$ and $\frac{dS(d-a)}{ad-bc}$

Example 17: Annual income of A and B is in the ratio of 5:4 and their annual expenses bear a ratio of 4:3. If each of them saves `500 at the end of the year, then find their annual income.

Solution: We have, a:b = 5:4, c:d = 4:3 and S = 500.

Annual income of A = $\frac{aS(d-c)}{ad-bc}$

$= \frac{5\ x\ 500\ x\ (3-4)}{(5x3-4x4)}$

= ₹2500

And annual income of B = $\frac{dS(d-c)S}{ad-bc}$

$= \frac{4\ x500\ x(3-4)}{(5x3-4x4)}$

= ₹2000

Example 18: The incomes of Mohan and Sohan are in the ratio 7:2 and their expenditures are in the ratio 4:1. If each saves `1000, find their expenditures.

Solution: We have, a:b = 7:2, c:d = 4:1 and S = 1000.

∴ Mohan's expenditure = $\frac{cS(b-a)}{ad-bc}$

$= \frac{4\ x\ 1000\ x\ (2-7)}{(7\ x\ 1-2x4)}$

= ₹2000

Sohan's expenditure = $\frac{dS(b-a)}{ad-bc}$

$= \frac{1\ x\ 1000x(2-7)}{(7x1-2x4)}$

₹5000

8) (a) If in a mixture of x litres of two liquids A and B, the ratio of liquids A and B is a:b, then the quantity of liquid B to be added in order to make this ratio.

C:d is $\frac{x(ad-bc)}{c(a+b)}$

Quantity of liquid A in the mixture = $\frac{ax}{a+b}$. Quantity of liquid B in the mixture = $\frac{bx}{a+b}$

Let/litres of liquid B to be added in order to make this ratio as c:d

Then $\frac{ax}{a+b} : \frac{bx}{a+b}$ + l = c:d

Or $\frac{ax}{a+b} : \frac{bx+l(a+b)}{a+b}$ = c:d

Or, : $\frac{ax}{bx+l(a+b}$ = C/d or axd = bcx +cl(a+b)

L= $\frac{x(ad-bc)}{(a+b)c}$

(b) In a mixture of two liquids A and B, the ratio of liquids A and B is a:b. If on adding x litres of liquid B to the mixture, the ratio of A to B becomes a:c, then in the beginning the quantity of liquid A in the mixture was ax/c-b− litres and that of liquid B was bx /c b − litres.

Solution

Let the quality of mixture be M liters

Let the quantity of mixture be M litres.

Then, the quantity of liquid A = $\frac{aM}{a+b}$ litres

and the quantity of liquid B = $\frac{bM}{a+b}$ litres.

If x litres of liquid B is added, then

$\frac{aM}{a+b}$: $\frac{bM}{a+b}$ + x= a:c

Or, $\frac{aM}{a+b}$: $\frac{bM+x(a+b)}{a+b}$ = a:c

or $\frac{aM}{bM+x(a+b)} = \frac{a}{c}$ or cM = bM +x(a+b)

Or M = $\frac{x(a+b)}{c-b}$

∴ quantity of liquid A = $\frac{ax(a+b)}{(c-b)(a+b)}$

= $\frac{ax}{c-b}$ litres

And, quantity of liquid B = $\frac{bx(a+b)}{(c-b)(a+b)}$

= $\frac{bx}{c-b}$ litres

Example 19: 729 ml of a mixture contains milk and water in the ratio 7:2. How much more water is to be added to get a new mixture containing milk and water in the ratio of 7:3. Solution: Here, x = 729, a:b = 7:2 and c:d = 7:3. ∴ The quantity of water to be added

= $\frac{x(ad-bc)}{c(a+b)}$

= $\frac{729\,x\,(7x\,3-2\,x7)}{7(7+2)}$

= 81 ml

Example 20: A mixture contains alcohol and water in the ratio of 6:1. On adding 8 litres of water, the ratio of alcohol to water becomes 6:5. Find the quantity of water in the mixture.

Solution: We have, a:b = 6:1, a:c = 6:5 and x = 8. ∴ The quantity of water in the mixture

= $\frac{bx}{c-b} = \frac{1\,x\,8}{5-1}$

= 2 litres

9)When two ingredients A and B of quantities q_1 and q_2 and cost price/unit c_1 and c_2 are mixed to get a mixture c having cost price/unit cm, then

a) $\frac{q_1}{q_2} = \frac{c_2-c_m}{c_m-c_1}$

b) $c_m = \frac{c_1 x q_1 + c_2 x q_2}{q_1+q_2}$

Example 21: In what ratio the two kinds of tea must be mixed together, one at ₹9 per Kg and another at ₹15 per Kg, so that mixture may cost ₹10.2 per Kg?

Solution: We have, c_1 = 9, c_2 = 15, cm = 10.2

∴ $\frac{q_1}{q_2} = \frac{c_2-c_m}{c_m-c_1} = \frac{15-10.2}{10.2-9}$

$=\frac{4.8}{1.2}$

$=\frac{4}{1}$

Thus, the two kinds of tea are mixed in the ratio 4:1

Example 22: In a mixture of two types of oils O_1 and O_2, the ratio O_1:O_2 is 3:2. If the cost of oil O_1 is ₹4 per litre and that of O_2 is ₹9 per litre, then find the cost per litre of the resulting mixture. Solution: We have, $q_1 = q_2 = 2$, $c_1 = 4$ and $c_2 = 9$. ∴ The cost of resulting mixture

$= \frac{c_1 x q_1 + c_2 x q_2}{q_1+q_2} = \frac{4x3+9\,x2}{3+2}$

$= \frac{30}{5}$

= ₹ 6

(a) If a mixture contains two ingredients A and B in the ratio a:b, then

percentage of A in the mixture = $\frac{a}{a+b}$x100% and percentage of B in the mixture $\frac{b}{a+b}\times$ 100%

(b) If two mixtures M_1 and M_2 contain ingredients A and B in the ratios a:b and c:d, respectively,

then a third mixture M_3 obtained by mixing M_1 and M_2 in the ratio x:y will contain

$\left[\frac{\frac{a*x}{a+b}+\frac{c*y}{c+d}}{x+y}\right]$ x 100% ingredient A , and

$\left[100\% - \left\{\frac{\frac{ax}{a+b}+\frac{cy}{c+d}}{x+y}\right\}\right]$

Or $\left[\frac{\frac{bx}{a+b}+\frac{dy}{c+d}}{x+y}\right]$ x 100% ingredient B

Example 23: Two alloys contain silver and copper in the ratio 3:1 and 5:3. In what ratio the two alloys should be added together to get a new alloy having silver and copper in the ratio of 2:1? Solution: We have, a:b = 3:1, c:d = 5:3 Let the two alloys be mixed in the ratio x:y. Then, percentage quantity of sliver in the new alloy

$= \left[\frac{\frac{a*x}{a+b}+\frac{c*y}{c+d}}{x+y}\right]$ x 100%

$= \left[\frac{\frac{3x}{4}+\frac{5y}{8}}{x+y}\right]$ x 100%

$= \frac{6x+5y}{8(x+y)}$ x 100% ..(1)

Since, the ratio of silver and copper in the new alloys is 2:1.

∴ percentage quantity of sliver in the new alloy

$= \frac{2}{2+1}$ x 100% $= \frac{200}{3}$% ….(2)

From (1) and (2), we get

$= \frac{6x+5y}{(x+y)} = \frac{2}{3}$ or 18x + 15 y = 16x + 16 y

or, 2x = y or, x:y = 1:2. Hence, the two alloys should be mixed in the ratio 1:2.

Summary

The key concepts learned from this unit are: -

- We have learnt how to solve to comparison of two quantities
- We have learnt how to solve Proportion
- We have learnt how to compare in a given mixture
- We have learnt how to solve and find Fraction

- We have learnt how to solve Equivalent ratios
- We have learnt how to solve Mean Proportional

Keywords

- Ratio
- Proportion
- Comparison
- Fraction
- Percent
- Equivalent ratios
- Mean Proportional

Self Assessment

1. Divide $200 in the ratio 3 : 2. What is the result?
   A. $140 : $60
   B. $80 : $120
   C. $300 : $200
   D. $120 : $80

2. The ratio of men to women at a party was 6 : 7, respectively.
   There were 28 women. Calculate the number of people at the party.
   A. 24
   B. 28
   C. 52
   D. 104

3. $x is divided among three boys, Ryan, Keith and Andrew, in the ratio 2 : 3 : 7, respectively.
   If Andrew gets $15 more than Keith, what is the value of x ?
   A. $27
   B. $45
   C. $57
   D. $180

4. A sum of J$300 is divided in the ratio 2 : 3.
   Calculate the amount of the LARGER share.
   A. J$120
   B. J$150
   C. J$180
   D. J$240

5. A metal is made from copper, zinc and lead in the ratio 13 : 6 : 1. The mass of the zinc is 90 kg. Calculate the mass of the metal.
   A. 195 kg
   B. 210 kg
   C. 285 kg
   D. 300 kg

6. The ratio of Rs 8 to 80 paise is
   A. 1 : 10
   B. 10 : 1
   C. 1: 1
   D. 100 : 1

7. The length and breadth of a steel tape are 10m and 2.4cm, respectively. The ratio of the

length to the breadth is
A. 5 : 1.2
B. 25: 6
C. 625: 6
D. 1250: 3

8. Find the missing number in the box in the following proportion:: 8 :: 12 : 32
A. 2
B. 3
C. 4
D. 5

9. State whether the given statement are true or false: 12 : 18 = 28 : 56

10. Fill in the blanks:
If two ratios are ______, then they are in proportion.

11. The first, second and fourth terms of a proportion are 16, 24 and 54 respectively. Then the third term is:
A. 36;
B. 28;
C. 48;
D. 32

12. If x, y and z are in proportion, then:
A. x : y : : z : x;
B. x : y : : y : z;
C. x : y : : z : y;
D. x : z : : y : z

13. Length and width of a field are in the ratio 5 : 3. If the width of the field is 42 m then its length is:
A. 100 m;
B. 80 m;
C. 50 m;
D. 70 m

14. The value of m, if 3, 18, m, 42 are in proportion is:
A. 6;
B. 54;
C. 7;
D. none of these

15. The ratio of 1 hour to 300 seconds is:
A. 1 : 12;
B. 12 : 1;
C. 1 : 5;
D. 5 : 1

16. State whether the given statements are true or false: 25 persons : 130 persons = 15kg : 78kg

Answers for Self Assessment

1.	D	2.	C	3.	B	4.	C	5.	D
6.	D	7.	B	8.	D	9.	B	10.	Equal/Same
11.	A	12.	B	13.	D	14.	C	15.	B
16.	A								

Review Questions

1. You jog 3.6 miles in 30 minutes. At that rate, how long will it take you to jog 4.8 miles?

2. You earn $33 in 8 hours. At that rate, how much would you earn in 5 hours?

3. An airplane flies 105 miles in ½ hour. How far can it fly in 1 ¼ hours at the same rate of speed?

4. What is the cost of six filters if eight filters cost $39.92?

5. If one gallon of paint covers 825 sq. ft., how much paint is needed to cover 2640 sq. ft.?

6. A map scale designates 1″ = 50 miles. If the distance between two towns on the map is 2.75 inches, how many miles must you drive to go from the first town to the second?

7. Bob is taking his son to look at colleges. The first college they plan to visit is 150 miles from their home. In the first hour they drive at a rate of 60 mph. If they want to reach their destination in 2 ½ hours, what speed must they average for the remainder of their trip?

8. Four employees can wash 20 service vehicles in 5 hours. How long would it take 5 employees to wash the same number of vehicles?

9. These two figures are similar. Use a proportion to find the length of side n.

20 m 12 m 30 m n

Set up a proportion to solve each word problem. (You may use the back to show your work)

10. Zach can read 7 pages of a book in 5 minutes. At this rate, how long will it take him to read the entire 175 page book? (Convert your final answer to hours and minutes)

11. Shannon's bicycle travels 50 feet for every 3 pedal turns. How many pedal turns are needed to travel one mile (1 mile = 5,280 feet)?

12. A cookie recipe for 5 dozen cookies calls for 4 cups of flour. How much flour is needed to make 90 cookies?

13. A man can make 12 birdhouses in 8 hours. How many hours will it take him to make 30 birdhouses?

Further Readings

1. Quantitative Aptitude For Competitive Examinations By Dr. R S Aggarwal, S Chand Publishing
2. A Modern Approach To Verbal & Non-Verbal Reasoning By Dr. R S Aggarwal, S Chand Publishing
3. Magical Book On Quicker Maths By M Tyra, Banking Service Chronicle
4. Analytical Reasoning By M.K. Pandey, Banking Service Chronicle